Lierda NT26-KCN E Series

Hardware Reference Design Manual

Version: Rev1.0

Date: 26/05/12

Status: Released

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Document Revision History

Document version	Change Date	Reviser	Reviewer	Change content
Rev1.0	26-05-12	CWY	SLY、YMX	Initial version

Safety Instructions

Users are responsible for complying with the relevant regulations and specific usage environment laws regarding wireless communication modules and devices in other countries. By adhering to the following safety principles, personal safety can be ensured, and potential damage to products and the working environment can be prevented. Our company does not assume responsibility for any losses resulting from customers' failure to comply with these regulations.

Road safety first! When driving, do not use handheld mobile terminal devices unless they have a hands-free function. Please park before making a call!

Please turn off mobile terminal devices before boarding. The wireless functions of mobile terminals are prohibited from being activated on the plane to prevent interference with the aircraft's communication system. Ignoring this instruction may compromise flight safety and even violate the law.

When in hospitals or healthcare facilities, pay attention to whether there are restrictions on the use of mobile terminal devices. RF interference may cause malfunction of medical equipment, so it may be necessary to turn off mobile terminal devices.

Mobile terminal devices do not guarantee effective connection in all circumstances, such as when the mobile terminal device has no balance or the SIM card is invalid. In case you encounter the above situations during an emergency, please remember to use emergency calls, ensuring that your device is powered on and located in an area with sufficient signal strength.

Your mobile terminal device will receive and transmit radio frequency signals when powered on, which may cause radio frequency interference when near televisions, radios, computers, or other electronic devices.

Please keep mobile terminal devices away from flammable gases. When you are near gas stations, oil depots, chemical plants, or explosion operation sites, please turn off mobile terminal devices. Operating electronic devices in any potentially explosive areas poses safety risks.

Applicable Module Selection

Serial Number	Module model	Supported frequency bands	Explanation
1	NT26K2E0-0G	Band1/3/5/8/ 34/38/39/40/41*	VDD_EXT power down in sleep mode
2	NT26K2E1-0G	Band1/3/5/8/ 34/38/39/40/41*	VDD_EXT sleep mode power down, module only supports single Beidou version.

Contents

1Introduction

This document provides application reference circuits for the Lierda NT26-KCN E series Cat.1 module, as well as some considerations in circuit design.

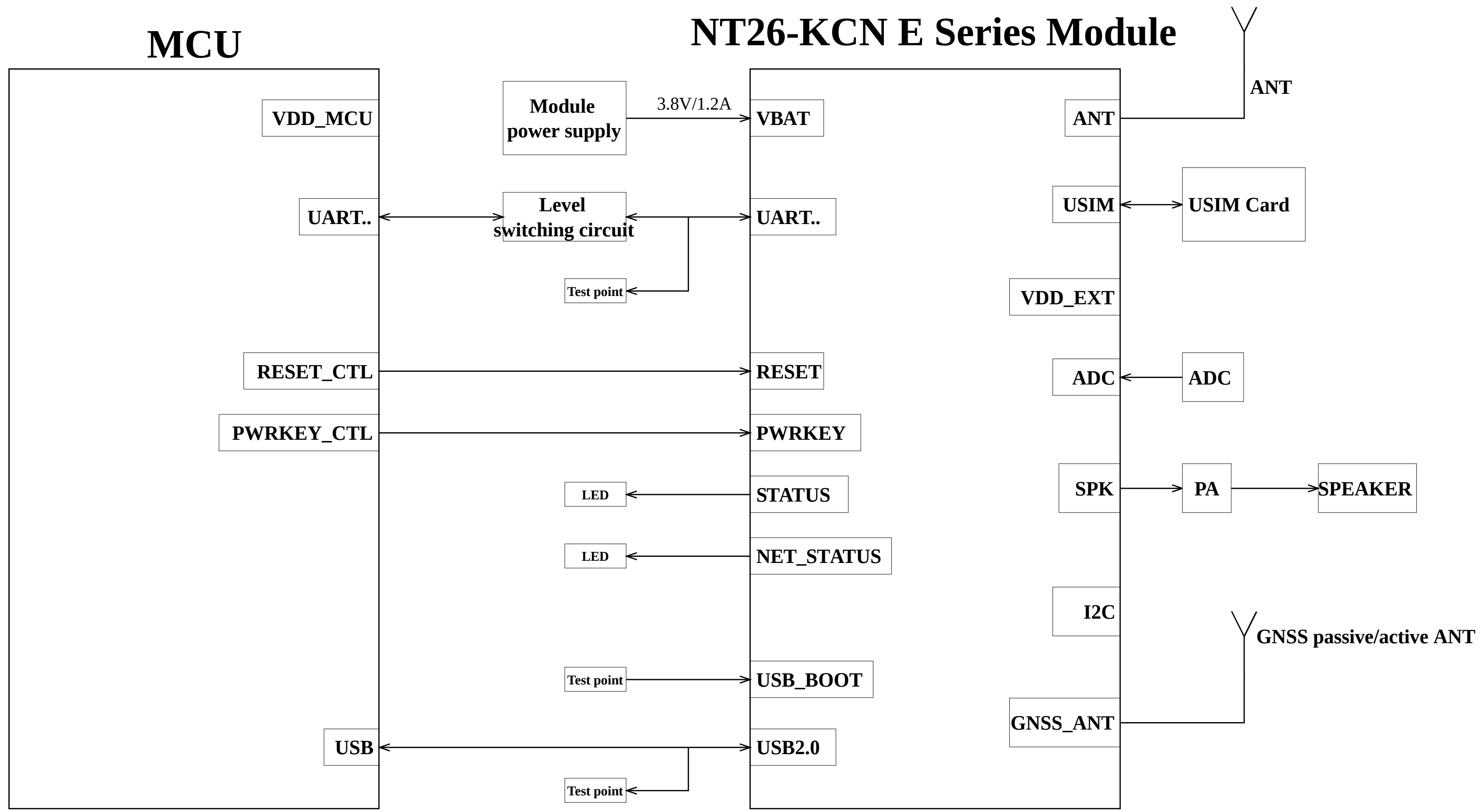
This document can help users quickly understand the peripheral hardware circuit of the module. Combined with other relevant documents, users can quickly grasp the application methods of the Cat.1 module.

2Reference Design

This chapter is the hardware reference design for the NT26-KCN E series, mainly including the design of power supply, SIM card, serial port, control signals, and peripheral application interfaces.

2.1Schematic Diagram

The following is the reference design schematic for the NT26-KCN E series module, and this design is for reference only.



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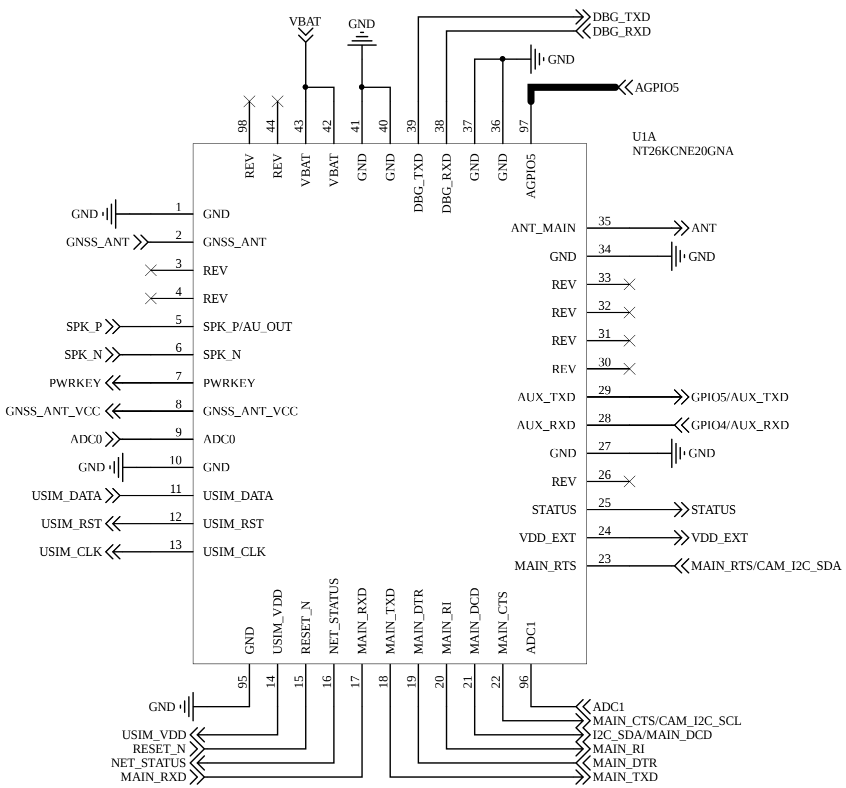

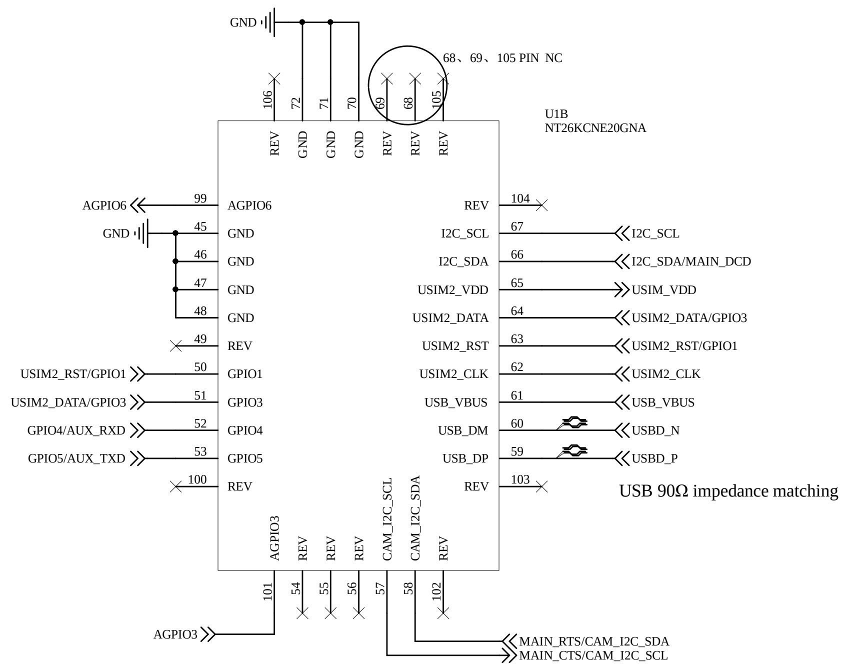

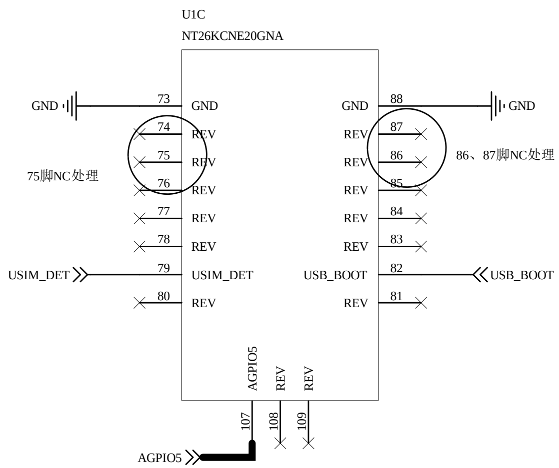

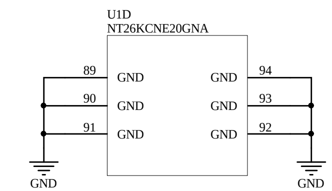

Comment：

1.USIM2_RST and GPIO1 are shared and cannot be used simultaneously.
2. USIM2_DATA and GPIO3 are shared and cannot be used simultaneously;
3. CAM_I2C_SCL and MAIN_CTS are shared and cannot be used simultaneously.
4. CAM_I2C_SDA and MAIN_RTS are shared and cannot be used simultaneously.
5. I2C_SDA and MAIN_DCD are shared and cannot be used simultaneously.
6. USB_BOOT and SPK_P are shared resources and cannot be used simultaneously.
7. AUX_RXD is mapped to GPIO4 and cannot be used simultaneously.
8. AUX_TXD is shared with GPIO5 and cannot be used simultaneously.
9. MAIN_RTS and CAM_I2C_SDA are shared and cannot be used simultaneously;
10. MAIN_CTS and CAM_I2C_SCL are shared and cannot be used simultaneously;
11. These modules (AGPIO3, AGPIO5, AGPIO6) are used internally within the module and cannot be externally connected for use.

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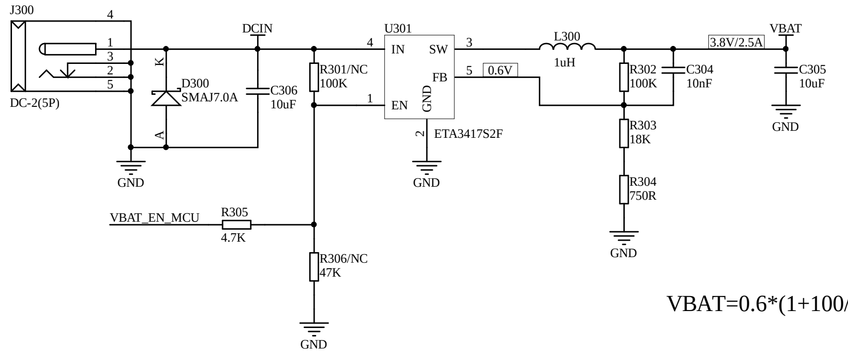
DC-DC power supply scheme

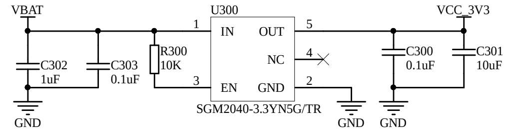


VBAT=0.6*(1+100/18.75)=3.80V

Comment：

1. When the input power supply voltage differs significantly from the module's power supply voltage, choose DC-DC power supply.
2. This circuit supports a maximum input of 7V and the output current can reach 2.5A.
3. The selection of the D200 passive protection device should be based on the actual input voltage.
3. The DC-DC circuit can be regarded as the pre-stage voltage reduction circuit, and then an LDO circuit can be connected after it to improve the stability of the power suppl
4. In low-power consumption scenarios, if enable control is needed, attach R305, do not attach R301. Under no-load conditions, the maximum static power consumption of th
5. If enable control is not needed, attach R301 directly, do not attach R305.

LDO（Power for Codec）

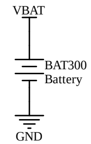


Comment：

1.The LDO outputs a 3.3V voltage with a current capacity of 250mA, and it mainly supplies power to the Codec chip.

2. This solution has an LDO static current of less than 1uA, tion scenarios. making it highly suitable for low-power consumption applica

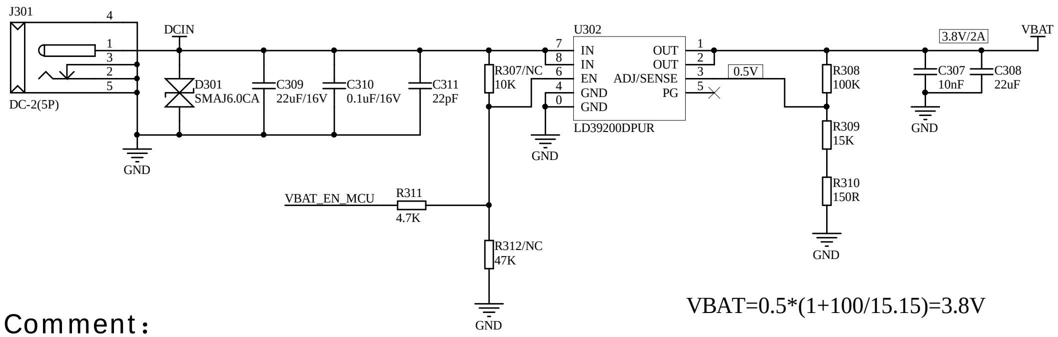
Battery-powered solution



Comment：

1.If battery power supply is chosen, the battery voltage should be between 3.3V and 4.5V, and it can be directly used for power supply.

LDO power supply scheme

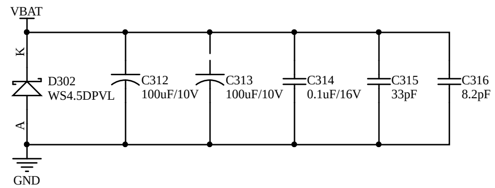


VBAT=0.5*(1+100/15.15)=3.8V

Comment：

1. When the input power supply voltage is not significantly different from the module's power supply voltage (for example, 5V), choose LDO power supply;
2. The maximum operating voltage of LDO supports 6V input and the output current can reach 2A;
3. The selection of the passive protection device D201 needs to be based on the actual input voltage;
4. In the low power consumption scenario of the entire machine, if enable control is needed, attach R311, do not attach R307. In no-load c
5. If no enable control is needed, attach R307 directly and do not attach R311.

Module VBAT power supply



Comment：

1.The input voltage range of VBAT is 3.3 to 4.5V.

2. The capacitor should be placed close to the power pins of the module.

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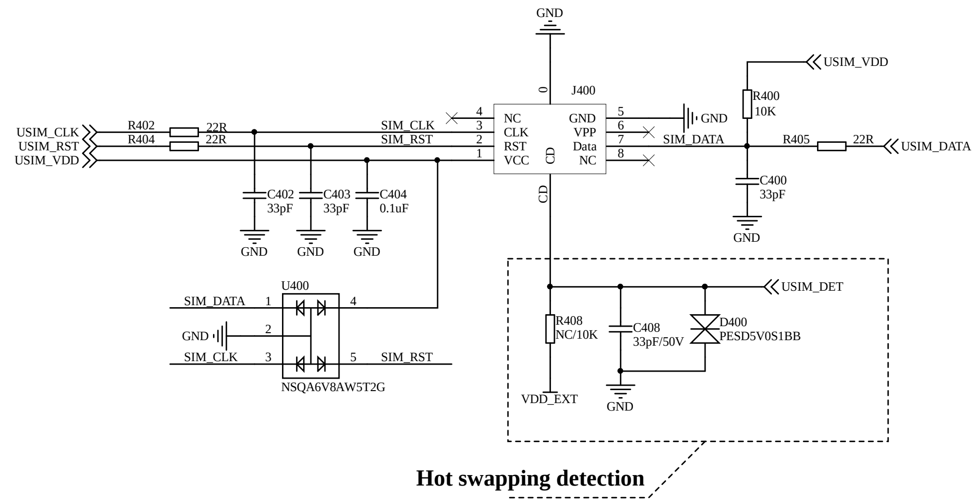
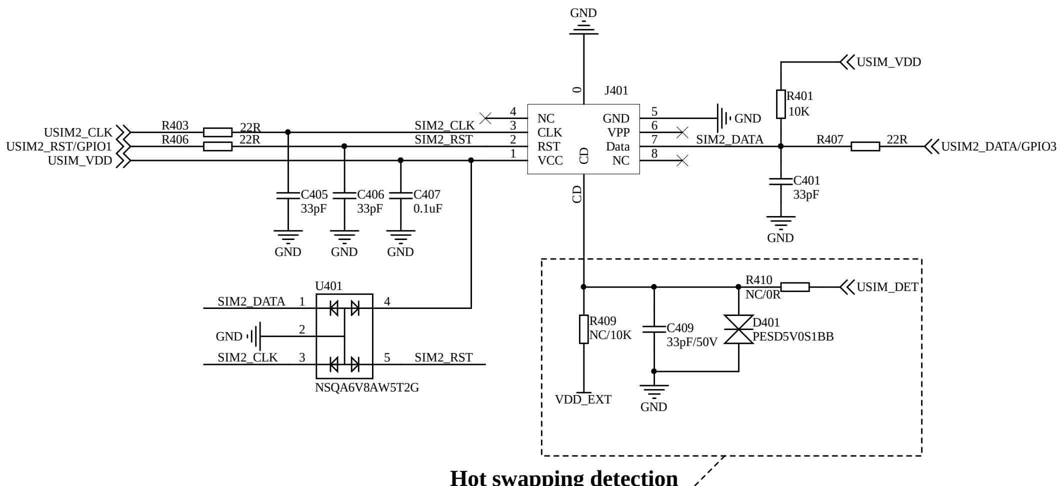
USIM card application circuit



USIM2 secondary card application circuit



The primary and secondary cards cannot be used simultaneously with USIM_DET.

Comment：

1. The filter capacitor of USIM_VDD should not exceed 1uF.
2. When using the plug-in card, the USIM wiring needs to add ESD protection devices, and the ESD device should be placed close to the SIM card slot, with the junction capacitance not exceedi
3. For USIM_DATA, it is recommended to externally connect a 10K pull-up resistor to USIM_VDD to enhance the anti-interference ability.
4. Between the USIM pin of the module and the card slot, a 22Ω resistor should be connected in series to suppress interference. Additionally, add a 33pF capacitor to filter out radio frequency in
5. Hot-plug detection is only applicable to SIM card slots with detection pins. If the hot-plug detection function is not required, the hot-plug detection part of the circuit can be removed.
6. SIM_DET is default high-active. High and low level detection can be configured. If this function is not needed, the module's hot-plug detection function can be turned off by cooperating with

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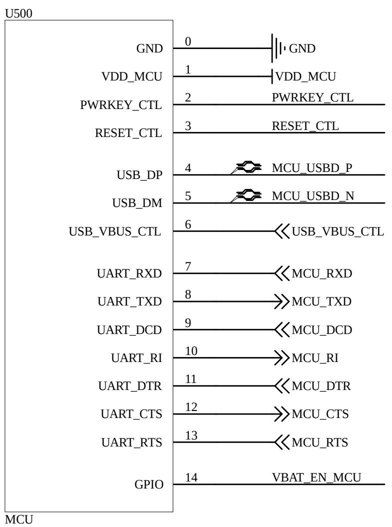
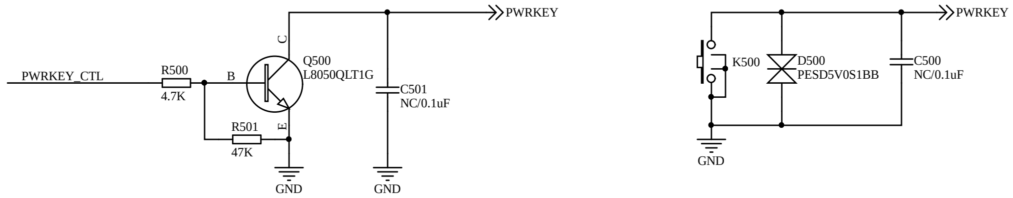
MCU interface



Power-on/off reference circuit (controlled by MCU or buttons; either option is acceptable)



Comment：

1.When using the MCU (IO: PWRKEY_CTL) to control the power-on/off pins of the module, it is preferable to use an open-collector drive circuit for control.

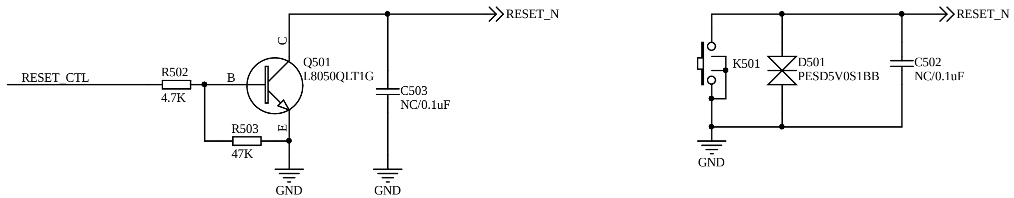
2. It is recommended to reserve a 0.1uF capacitor at the module pin position. By default, do not attach it.

Comment：

1. It is recommended to reserve a 0.1uF capacitor at the module pin position. By default, it is not to be attached.

2. Place an ESD device close to the button for electrostatic protection.

Reset reference circuit (controlled by MCU or key press, either option is acceptable)



Comment：

1.When using the MCU (IO: RESET_CTL) to control the reset pin of the module, it is preferable to use an open-collector drive circuit for control.

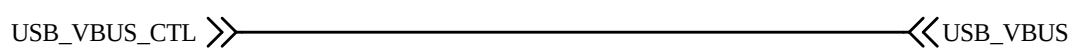
2. It is recommended to reserve a 0.1uF capacitor at the module pin position. By default, do not attach it.

Comment：

1.It is recommended to reserve a 0.1uF capacitor at the module pin position. By default, it is not to be attached.

2. Place an ESD device close to the button for electrostatic protection.

USB_VBUS Control Circuit

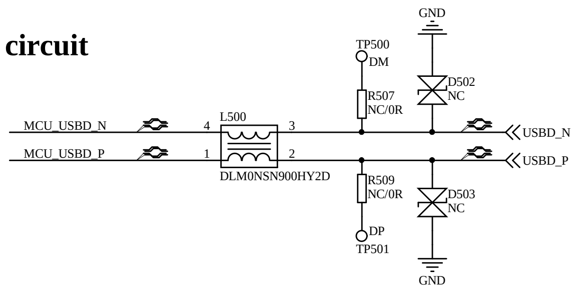


Comment：

1.This circuit can also wake up the module by controlling the USB_VBUS of the module through the USB_VBUS_CTL of the MCU.

2. If it is an external USB interface, USB_5V connects to USB_VBUS, and by simply inserting the USB cable, the module can be awakened.

USB circuit



Comment：

1.It is recommended to add a common-mode inductor to the USB cable between the MCU and the module to reduce EMI interference.

2. Common-mode inductors, resistors, and test points are placed close to the module.

3. USB cable control differential 90-ohm impedance;

4. For passive protection devices D502 and D503 for USB signal lines, it is recommended to select models with a capacitance of ≤ 0.6pF.

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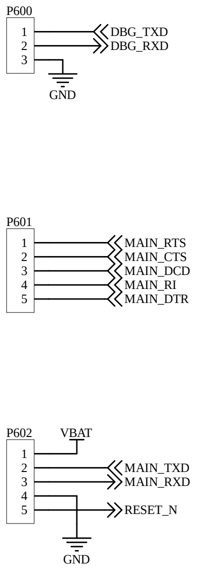
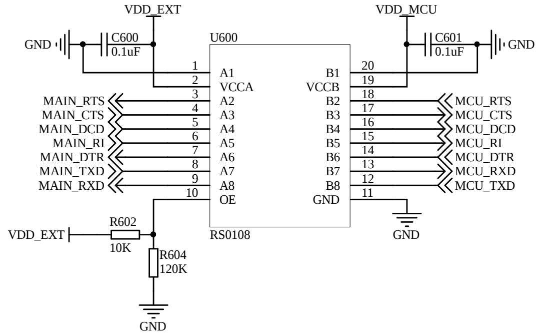
UART interface



Comment：

1. DBG_UART can be used to view the LOG data of the module.
   It is recommended to reserve test points.
2. MAIN_UART is used for data transmission, AT commands,
   and firmware upgrade. It supports hardware flow control
   and low-power wake-up. It is recommended to reserve test points.
3. When connecting between the MCU and the module UART,
   pay attention to the level matching issue.
4. MAIN_RI and MAIN_DTR belong to the constant power supply domain
   and can be configured to not lose power during sleep mode.

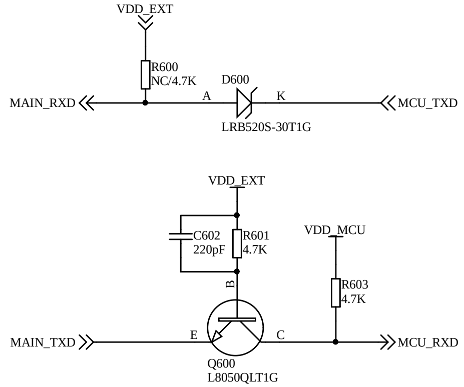
Level conversion circuit (chip solution)



Comment：

1. When using the level conversion chip circuit, it is necessary to note that VCCA ≤ VCCB.
   For details, please refer to the chip data sheet.
2. The chip enable pin OE is connected to the reference level VCCA. Pulling up OE enables bidirectional level conversion.
3. The rate of the level conversion chip: push-pull mode 20Mbps, open-collector mode 2Mbps.
4. For the hardware version with VDD_EXT in sleep mode without power supply, MAIN_RXD cannot use this circuit.
5. The MIAN_DCD pin and the I2C_SDA pin are mutually multiplexed pins and cannot be used simultaneously.

Level conversion circuit (discrete component solution)



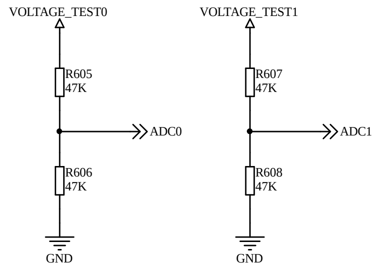
Comment：

1. When using the diode conversion circuit, a low-dropout (≤0.3V) Schottky diode should be selected;
2. The internal of the module MAIN_RXD has an upper pull-up resistor. R600 does not need to be attached.
   If there is an upper pull-up resistor inside MCU_RXD, R603 can be omitted from the welding;
3. The conversion circuits of MAIN_RTS and MAIN_DTR are similar to that of MAIN_RXD;
4. The conversion circuits of MAIN_CTS, MAIN_RI and MAIN_DCD are similar to that of MAIN_TXD;

Serial Port Level Conversion Circuit Description

These two level conversion circuits can be used independently or in combination.

ADC



Comment：

1. VOLTAGE_TEST represents the voltage to be tested;
2. In the internal direct connection mode (default):
The input voltage range for ADC0 to ADC1 is 0 to 1.05V;
3. In the internal voltage divider mode:
The input voltage range for ADC0 to ADC1 is 0 to 3.6V;

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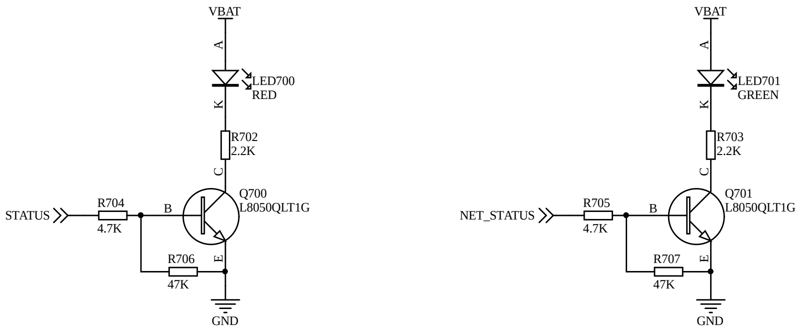
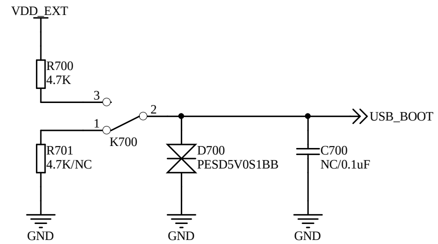
STATUS & NET_STATUS Reference Circuit



Comment：

1. STATUS is the module status indicator pin. After the module is powered on normally, STATUS outputs a high level.
2. NET_STATUS is the network status indicator pin, which will adjust the duration of high and low voltage levels according to the network status of the module.

USB_BOOT reference circuit

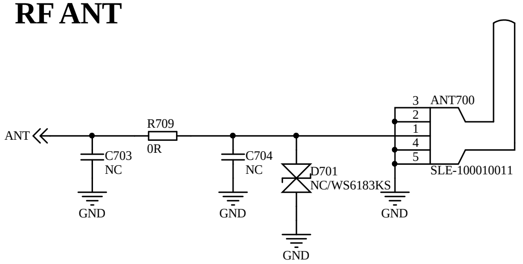


Comment：

1.Before powering on, pulling up USB_BOOT can enable the module to enter the emergency download mode after power-on, and then the firmware can be downloaded via the USB interface.

2. By default, please keep USB_BOOT disconnected.

RF ANT



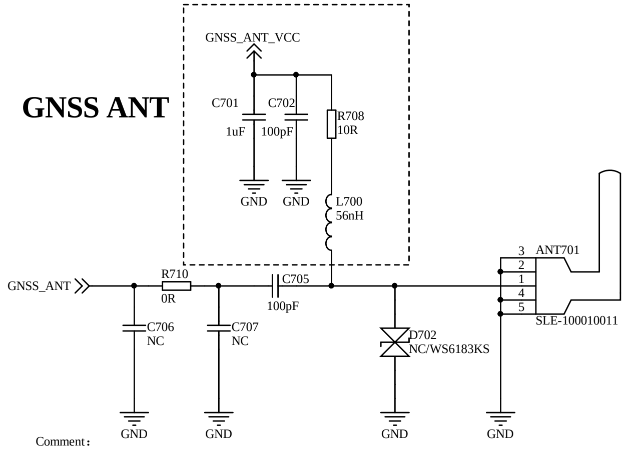
Comment：

1.It is recommended to reserve a π-type antenna matching circuit. The initial resistance is set to 0R, and the capacitor is left unconnected. The actual value will depend on the antenna.

2. To achieve better electrostatic protection performance, it is recommended to place the ESD components close to the antenna.

3. The parasitic capacitance of the ESD device at the antenna port is recommended to be no more than 0.5pF.

GNSS ANT



Comment：

1.If an active antenna is used, it can be powered by the module GNSS_ANT_VCC power supply, with the VCC power range being 3.3V.

2. If an inductive antenna is used, the internal control of the GNSS antenna power supply will be disconnected. The NC terminals of C701, C702, R708, and L700 can be left unconnected.

3. Depending on the antenna type, especially those prone to generating static electricity, in order to prevent damage to the module caused by static electricity, TVS diodes (≤ 0.5pF) or inductors (≥ 47nH) can be reserved near the antenna port.

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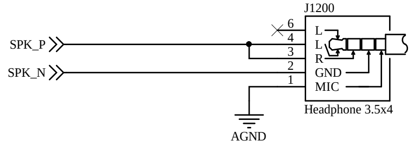
SPK_P/N Headphone Circuit

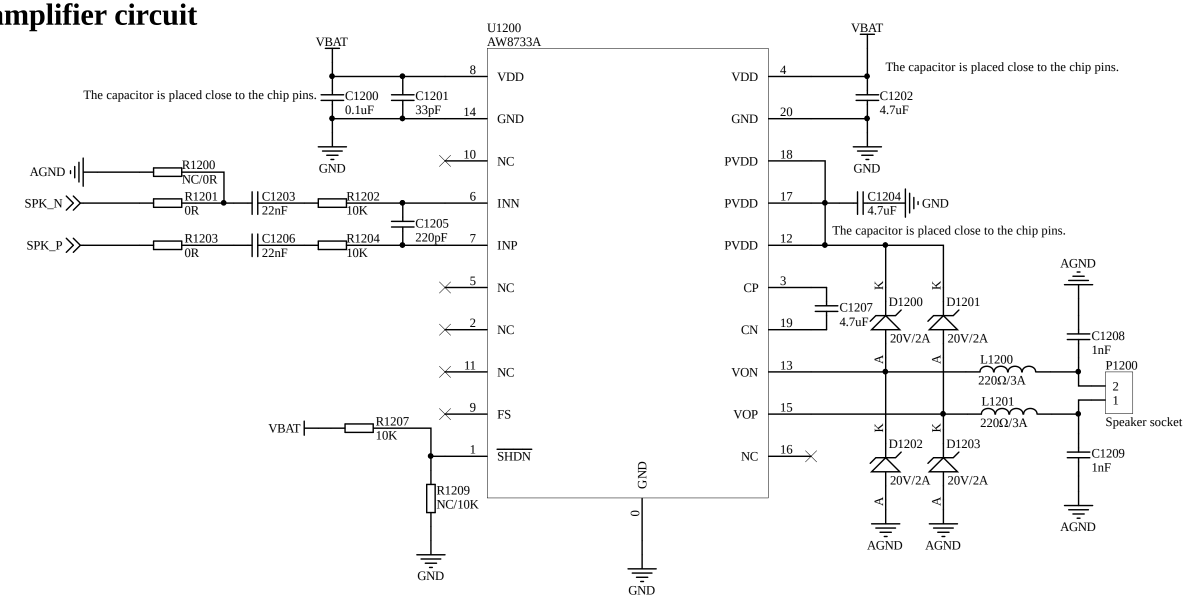


The headphone jack is connected according to the specific model.

Comment：

1. The output signal lines of SPK_P/SPK_N need to be grounded and kept away from interference sources.
2. The audio output power of the NT26-KCN E series module is limited. There are significant differences in impedance and sensitivity among various headphones.
For high-resistance and low-sensitivity headphones, the volume may be relatively weak. Users need to choose the appropriate headphones based on actual applications.

Audio power amplifier circuit



Comment：

1. PIN4 and PIN8 (VDD) must be connected together during layout, and all should be connected to VABT.
2. PIN12, PIN17, and PIN18 (PVDD) must be connected together during layout.
3. The input capacitors and resistors should be placed as close as possible to the INN and INP pins of the chip, and the input lines should run parallel to suppress noise coupling.
4. The beads and capacitors should be placed near the VON and VOP pins of the chip, and the output line from the chip to the speaker should be as short and thick as possible.
5. When only one audio SPK_P or SPK_N outputs independently, PIN6 is selected for grounding, and PIN7 is selected based on the output port of the module.

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3 Relevant documents and terminology abbreviations

The following related documents provide the names of the relevant documents, and the specific version shall be subject to the latest release.

Table 3-1 Related Documents

Serial Number	Document Name	Comment
[1]	Lierda NT26-KCN E Series (New Version)_Hardware Design Manual	

Table 3-2 Terms Abbreviation

Abbreviation	Full English name	Full name in Chinese.
ANT	Antenna	Antenna
DBG	Debug	Debugging
DC-DC	Direct Current - Direct Current	DC converter
GPIO	General-purpose input/output	General Input and Output
LDO	Low Dropout Regulator	Low Dropout Linear Regulator
MCU	Mirco Controller Unit	Microcontroller Unit
NB-IoT	Narrow Band Internet of Things	Narrowband Internet of Things
PSM	Power Saving Mode	Energy-saving mode
RF	Radio Frequency	RF
RX	Receive	Receive
TVS	Transient Voltage Suppressor	Transient suppression diode
TX	Transmit	Send
UART	Universal Asynchronous Receiver & Transmitter	Universal Asynchronous Receiver and Transmitter

USIM	Universal Subscriber Identification Module	General User Identification Module
GNSS	Global Navigation Satellite System	Global Navigation Satellite System